

Lecture 2: Planet Formation 101 & Orbital Dynamics

Introduction to Planetary Science

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Learning Objectives

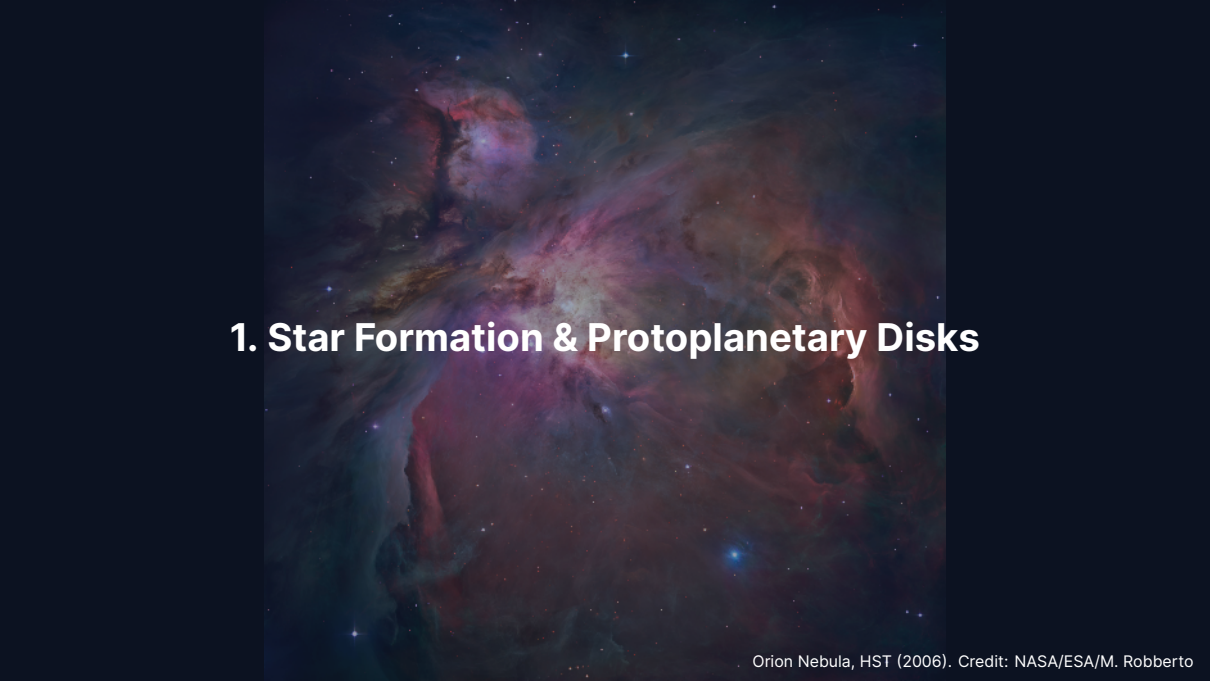
By the end of this lecture, you will be able to:

- Describe the stages of planet formation from molecular cloud to planets
- Explain how dust grows into planetesimals and planets
- Apply Kepler's laws and the vis-viva equation to orbital problems
- Discuss orbital resonances, tidal forces, and planetary migration

Recap: Lecture 1

- Three driving questions: formation, habitability, life
- The IAU planet definition and the Pluto debate
- Solar system architecture: inner rocky, outer gas/ice
- Derived $M_{\odot}$ from Kepler's third law
- Comparative planetology as methodology

Today: How did all of this come to be?

A detailed Hubble Space Telescope image of the Orion Nebula, showing vibrant red and blue gas clouds and numerous stars. The nebula's structure is complex, with various filaments and knots of gas. The background is a deep black space filled with distant stars.

1. Star Formation & Protoplanetary Disks

Orion Nebula, HST (2006). Credit: NASA/ESA/M. Robberto

From Molecular Cloud to Star

- Stars form from cold ($T \sim 10\text{--}20\text{ K}$), dense molecular clouds
- Collapse triggered when self-gravity overcomes thermal pressure
- Critical mass for collapse, the **Jeans mass**:

$$M_J \sim \frac{c_s^3}{\sqrt{G^3 \rho}}$$

- For typical conditions: $M_J \sim 1\text{--}10 M_\odot$
- Angular momentum conservation $\Rightarrow$ **disk** forms around the protostar

Jeans Instability: When Gravity Wins

Derivation. Compare gravitational free-fall time to sound-crossing time.

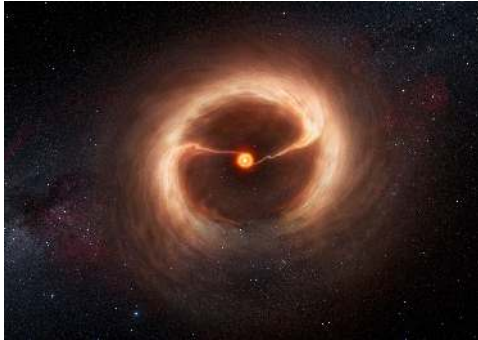
- Free-fall time: $t_{\text{ff}} = \sqrt{3\pi/(32 G\rho)}$
- Sound-crossing time across scale L : $t_{\text{sc}} = L/c_s$
- Collapse requires $t_{\text{ff}} < t_{\text{sc}}$, giving:

$$L > L_J = c_s t_{\text{ff}} \sim \frac{c_s}{\sqrt{G\rho}}$$

- For typical cloud conditions ($T = 10$ K, $n_{\text{H}_2} = 10^4 \text{ cm}^{-3}$): $L_J \sim 0.1$ pc, $M_J \sim 1\text{--}10 M_\odot$
- Free-fall time $\sim 10^5$ yr
- Fragmentation: if the cloud contains multiple Jeans masses, it can split into many protostars

Source: Jeans (1902); Shu (1977)

Protoplanetary Disk Structure



Credit: ESO/L. Calçada, CC BY 4.0

- Flattened rotating structure, ~few AU to ~hundreds AU
- Total mass $\sim 0.01\text{--}0.1 M_{\odot}$
- **Temperature gradient:** $>1500\text{ K}$ (inner) to $<30\text{ K}$ (outer)
- **Dust-to-gas ratio:** $\sim 1:100$ (by mass)
- Surface density: $\Sigma(r) \propto r^{-p}$

Vertical Structure and Flaring

The disk is not razor-thin: it has a finite vertical scale.

- Hydrostatic balance gives a Gaussian density profile with scale height H
- $H/r = c_s/v_K$ where c_s is sound speed and v_K is Keplerian speed
- Typical $H/r \sim 0.03\text{--}0.1$, increases outward (flared disk)
- Dust **settles** toward the midplane under vertical gravity
- Settling creates a denser dust layer ($H_{\text{dust}}/H_{\text{gas}} \sim 0.1$)
- Once the midplane dust-to-gas ratio approaches unity, instabilities take over

Dust settling is the first step toward planetesimal formation: it creates the conditions where streaming instability can act.

The Minimum-Mass Solar Nebula

If we spread the solar system's planets back into a disk, what mass would we need?

- Augment each planet's mass by solar H/He to match solar composition
- Spread over an annulus centred on each planet
- Result: **Minimum-Mass Solar Nebula (MMSN)**, Hayashi (1981)
- Total mass: $\sim 0.01\text{--}0.02 M_{\odot}$
- Surface density: $\Sigma_{\text{MMSN}}(r) \approx 1700 (r/\text{AU})^{-3/2} \text{ g/cm}^2$
- A convenient scaling, but modern observations favour more massive disks for Jovian cores

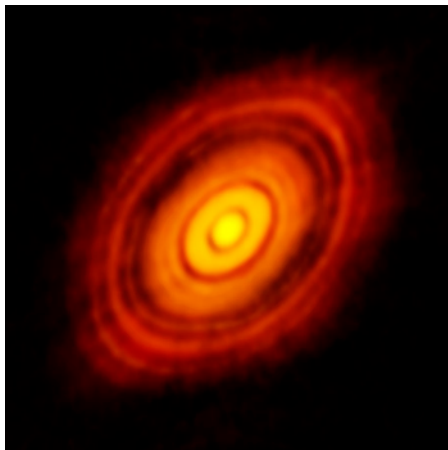
Source: Hayashi (1981); Desch (2007) for revisions

The Snow Line

- **Snow line** = distance where water condenses as ice (~150–170 K)
- In our solar system: ~2.5–3.5 AU (between Mars and Jupiter)
- Beyond the snow line:
 - Solid material ~3× more abundant (ice added to rock/metal)
 - Enables faster growth of planetary cores
- Other ice lines at greater distances: CO_2 , CO , N_2 , NH_3
- Creates a **compositional gradient** across the disk

The snow line explains the fundamental architecture of the solar system: small rocky planets inside, massive gas/ice giants outside.

ALMA: Disks Resolved

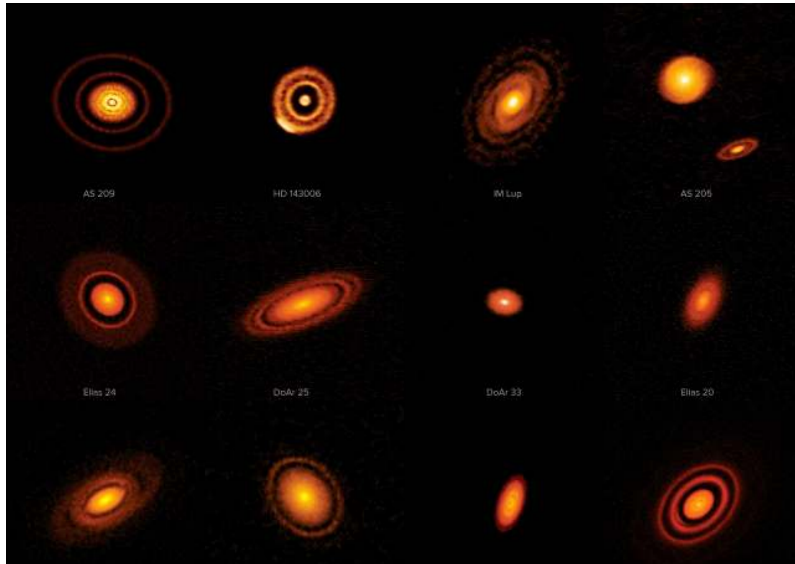


Credit: ALMA/ESO/NAOJ/NRAO (2014)

- **HL Tauri** (2014): ~1 Myr old star
- Concentric rings and gaps at AU resolution
- Gaps possibly carved by forming planets
- Disk substructure present even in very young systems

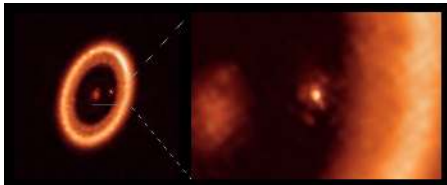
Planet formation begins earlier than previously thought.

DSHARP Survey: Substructure Is Universal



PDS 70: Planets Caught in the Act

In 2018–2019, VLT/SPHERE and ALMA directly imaged two protoplanets in the disk gap.



Credit: ALMA/ESO/NAOJ/NRAO (Benisty et al. 2021)

- PDS 70 b: $\sim 6 M_{\text{Jup}}$, ~ 20 AU
- PDS 70 c: similar mass, at ~ 35 AU
- H α emission: active accretion onto both planets
- Circumplanetary disk around PDS 70 c (2021)
- First direct evidence of giant planets forming in disk gaps
- Supports the DSHARP picture: gaps sculpted by embedded protoplanets

ALMA MAPS: Chemical Diversity in Disks

The Molecules with ALMA at Planet-Forming Scales (MAPS) survey (2021) resolved the chemical composition of 5 disks.

- Mapped the distributions of CO, HCO^+ , HCN, C_2H , CH_3CN , H_2CO , ...
- Showed that complex organic molecules are present at planet-forming distances
- Different chemistries at different radii: volatile C/O ratios vary with distance
- Snow lines for multiple species are directly observable
- Implications: the compositions inherited by forming planets depend on where in the disk they assemble

Source: Öberg et al. (2021), *ApJS*

Disk Lifetimes: A Hard Deadline

- Infrared surveys of young stellar clusters show disk dispersal:
 - ~80% of stars have disks at 1 Myr
 - <10% by 10 Myr
- **Median disk lifetime: 3–5 Myr**
- Dispersal driven by:
 - Viscous accretion onto the star
 - Photoevaporation by stellar UV/X-ray radiation
 - Magnetohydrodynamic winds

The disk is gone by ~3–5 Myr. Giant-planet **cores** must reach critical mass well before that, typically in the first 1–3 Myr, to capture substantial gas envelopes (Lambrechts & Johansen 2014).

A false-color image of the HL Tauri star system, showing a series of concentric rings of dust and gas surrounding the central star. The rings are composed of alternating layers of dust and gas, with the innermost ring being the most prominent. The color scale ranges from dark blue (low intensity) to bright yellow/white (high intensity).

2. From Dust to Planets

Grain Growth: Sticking and Settling

- Starting material: sub- μm dust grains (silicates, oxides, ices)
- Growth factor to planets: $\sim 10^{13}$ in size
- **Step 1: Sticking**
 - Low-velocity collisions ($\lesssim 1 \text{ m s}^{-1}$): van der Waals forces
 - Forms fractal aggregates up to mm–cm sizes
- **Step 2: Settling**
 - Dust settles to the disk midplane under vertical gravity
 - Creates a dense dust layer within the gas disk

Dust-Gas Coupling: The Stokes Number

How tightly is a particle coupled to the gas?

$$St = \tau_{\text{stop}} \Omega_K$$

where τ_{stop} is the friction stopping time and Ω_K is the Keplerian frequency.

$St \ll 1$ (small grains) perfectly coupled to gas; follow the flow

$St \sim 1$ (pebbles, cm-dm) partially decoupled; experience maximum radial drift

$St \gg 1$ (boulders, km) decoupled; move on Keplerian orbits

Particles with $St \sim 0.1$ are the “sweet spot” for pebble accretion: they drift efficiently yet can still be deflected by planetary gravity.

Source: Weidenschilling (1977); Johansen et al. (2014) in *Protostars & Planets VI*

Growth Barriers

Three barriers impede growth beyond cm-scale:

Bouncing barrier (~mm): Compacted aggregates bounce rather than stick

Fragmentation barrier (~cm): Collision velocities $>1\text{--}10\text{ m s}^{-1}$ shatter aggregates

Radial drift barrier (~metre): Gas headwind drains angular momentum
→ Metre-sized boulders spiral into the star in $\sim 100\text{--}1000\text{ yr}$

The metre-size problem

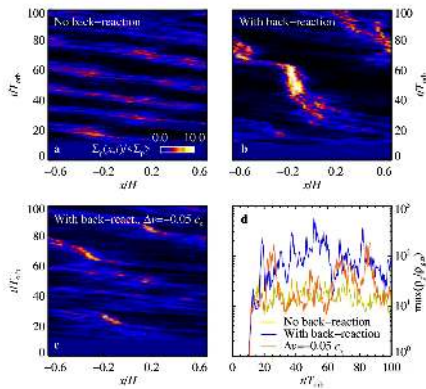
How does nature bridge the gap from cm to km?

Streaming Instability: Skipping the Barrier

- When local dust-to-gas ratio ≥ 1 (from settling + pressure bumps):
 1. Concentrated particles accelerate gas toward Keplerian speed
 2. Reduced headwind attracts more drifting particles
 3. **Positive feedback** → dense filaments and clumps
- Dense clumps collapse under self-gravity
- Directly form **planetesimals** (~50–500 km)
- Skips the problematic cm-to-metre range entirely

The streaming instability solves the radial drift barrier by collective effects: particles help each other avoid falling into the star.

Streaming Instability: A Simulation Snapshot



- **Panels a–c:** particle column density vs. radial position and time
- **Panel a:** no back-reaction on gas → only weak drift banding
- **Panel b:** with back-reaction → dense clumps form
- **Panel d:** maximum particle density rises to $>10^3$ background
- Clumps are dense enough for self-gravitating

Credit: Johansen et al. (2007), Nature 448, 1022 (arXiv:0708.3890)

Pebble Accretion: Fast Growth

- Once planetesimals exist, they sweep up mm–cm “pebbles”
- Gas drag deflects pebbles strongly toward growing body
- Effective cross-section $\gg$ geometric cross-section
- Enables growth to $\sim 10 M_{\oplus}$ within disk lifetime
- Fast enough to trigger gas accretion $\rightarrow$ giant planets

Planet formation sequence

Dust grains $\xrightarrow{\text{sticking}}$ aggregates $\xrightarrow{\text{streaming inst.}}$ planetesimals $\xrightarrow{\text{pebble acc.}}$ planetary cores

Gravitational Focusing

Why does the largest body in a swarm grow fastest?

$$\sigma_{\text{eff}} = \pi R^2 \left(1 + \frac{v_{\text{esc}}^2}{v_{\infty}^2} \right)$$

- Classical hard-sphere cross-section = πR^2
- Gravity of the body curves nearby trajectories inward
- Effective collision cross-section is enhanced by $(1 + v_{\text{esc}}^2/v_{\infty}^2)$
- When $v_{\text{esc}} \gg v_{\infty}$ (low random velocities), focusing is enormous
- Because $v_{\text{esc}} \propto \sqrt{M/R}$, the **largest** body focuses most effectively

Consequence: an initially small mass excess runs away into a dominant body.

Runaway & Oligarchic Growth

Runaway growth:

- Biggest body focuses best
- Grows fastest (positive feedback)
- Reaches Moon–Mars mass in $\sim 10^5$ yr
- Ends when the big body stirs up the swarm

Oligarchic growth:

- Winners stir up leftovers $\rightarrow$ focusing weakens
- A few dozen “oligarchs” dominate
- Each separated by ~ 5 – 10 Hill radii
- Isolation mass: ~ 0.01 – $0.1 M_{\oplus}$

Isolation Mass

How big can oligarchs get?

The **isolation mass** is reached when an oligarch has consumed all available planetesimals within its feeding zone:

$$M_{\text{iso}} \sim \frac{(4\pi B a^2 \Sigma_s)^{3/2}}{(3M_*)^{1/2}}$$

where $B \sim 5\text{--}10$ is the feeding zone width in Hill radii.

- Inner solar system ($\Sigma_s \sim 10 \text{ g/cm}^2$): $M_{\text{iso}} \sim 0.01\text{--}0.1 M_{\oplus}$
- Outer solar system (beyond snow line, $\Sigma_s \sim 30 \text{ g/cm}^2$): $M_{\text{iso}} \sim 10 M_{\oplus}$
- Explains why giant planet cores can reach critical mass outside the snow line but not inside

Core Accretion: Building Giant Planets

1. Solid core grows beyond the snow line (more solids available)
2. Core reaches **critical mass** $\sim 5\text{--}15 M_{\oplus}$
3. Gravity binds a gas envelope from the disk
4. Envelope contracts, cools $\rightarrow$ **runaway gas accretion**
5. Accretion stops when planet opens a gap or disk disperses

This explains:

- Jupiter & Saturn: formed beyond snow line, reached critical mass early
- Uranus & Neptune: reached critical mass too late, less gas captured

Alternative: Gravitational Instability

- If the disk is massive and cools rapidly:
- Gas disk fragments directly into giant planets
- Very fast: $\sim 10^3$ yr (vs. $\sim 10^6$ for core accretion)
- May work at large orbital distances (≥ 30 –50 AU)
- Debated for solar system giants

Two pathways

Core accretion:

bottom-up (dust $\rightarrow$ core $\rightarrow$ gas)

Grav. instability:

top-down (disk $\rightarrow$ clump $\rightarrow$ planet)

Planet–Planet Scattering

After the gas disk disperses, gravitational interactions among embryos become **chaotic**.

- Initially packed oligarchs cross each other's Hill radii
- Close encounters pump eccentricities on ~Myr timescales
- Orbits intersect $\Rightarrow$ collisions or ejections
- Survivors spiral toward stable separations (~10–20 mutual Hill radii)
- Damping by residual planetesimals circularises orbits over 10–100 Myr

Scattering is the dominant mechanism for producing the observed eccentricity distribution of exoplanets, including eccentric giant planets at intermediate separations.

Source: Rasio & Ford (1996); Chambers (2004); Lissauer et al. (2011)

The Giant Impact Phase



Credit: NASA/JPL-Caltech (artist's concept)

- After disk dispersal: embryos' orbits become chaotic
- ~10–100 Myr of collisions assemble final terrestrial planets
- **Moon-forming impact:** proto-Earth hit by Mars-sized Theia
- Melts the planet → magma ocean → core formation

Formation is not just assembly; it drives differentiation, outgassing, and sets the stage for all subsequent evolution (Lectures 3 & 4).

From Dust to Planets: The Full Sequence



Credit: Wikimedia, public

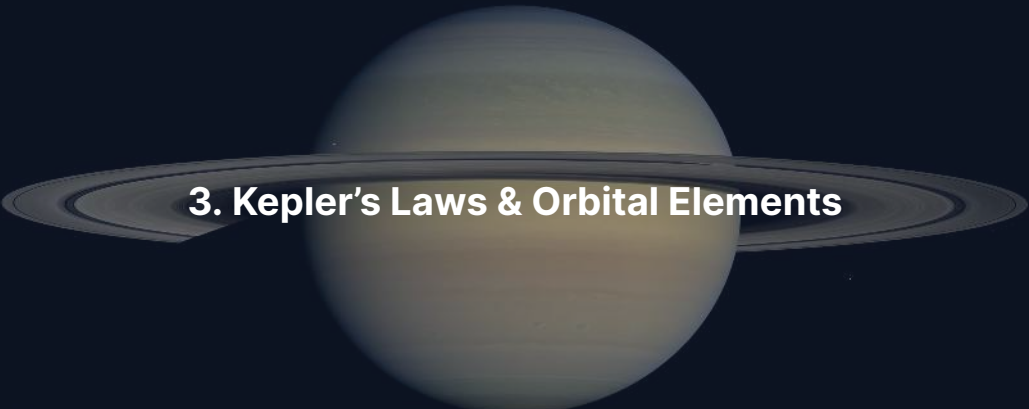
Four key stages, from top to bottom:

1. **Cloud collapse:** a cold, dense molecular cloud fragments and contracts under self-gravity
2. **Protoplanetary disk:** conservation of angular momentum flattens the infalling gas into a rotating disk
3. **Disk substructure:** dust settles and concentrates into rings; streaming instability builds planetesimals
4. **Mature system:** runaway and oligarchic growth, followed by giant impacts, produce the planets on stable orbits

Total timescale: ~100 Myr from first collapse to a settled planetary system.

Break



A large, pale yellow planet with a prominent ring system, centered in the frame against a dark blue background. The rings are composed of many thin, concentric bands of varying shades of grey and white. The planet's surface shows subtle horizontal banding.

3. Kepler's Laws & Orbital Elements

Kepler's Three Laws

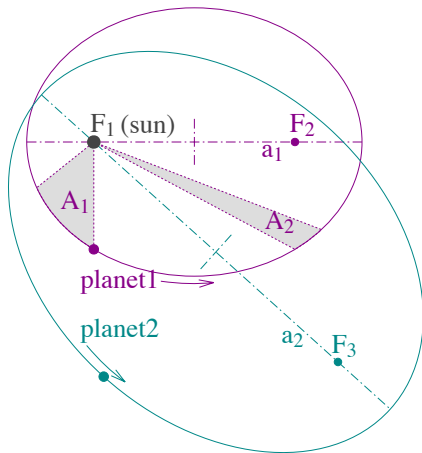
1. **Law of ellipses:** Each planet orbits the Sun on an **ellipse** with the Sun at one focus.
2. **Equal areas:** A line from planet to Sun sweeps **equal areas in equal times**.
⇒ faster at perihelion, slower at aphelion
3. **Harmonic law:**

$$P^2 = \frac{4\pi^2}{G(M_1 + M_2)} a^3$$

Connects period P to semi-major axis a and total mass.

Newton showed: **all three follow from universal gravitation.**

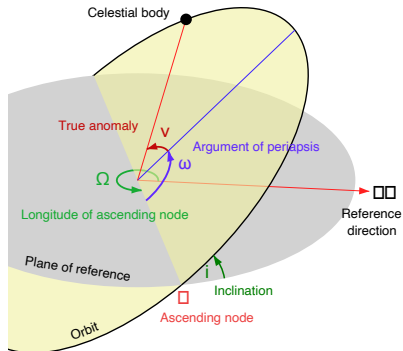
Visualising Kepler's Laws



Credit: Hankwang, CC BY 2.5

Elliptical orbit (left), equal areas (centre, shaded regions have equal area), and the period–semi-major axis relation (right).

Six Orbital Elements



Credit: Lasunncty, CC BY-SA 3.0

Element	Symbol	Controls
Semi-major axis	a	Size
Eccentricity	e	Shape
Inclination	i	Tilt
Long. asc. node	Ω	Twist
Arg. perihelion	ω	Orientation
True anomaly	v	Position

First five = orbit shape & orientation.
Sixth = where the body is now.

The Two-Body Problem

Two masses m_1, m_2 interacting through gravity:

- Reduces to **reduced mass** $\mu = m_1 m_2 / (m_1 + m_2)$ in central field of $M = m_1 + m_2$
- Total orbital energy:

$$E = \frac{1}{2} \mu v^2 - \frac{G m_1 m_2}{r}$$

- Fundamental result: energy depends **only on** a :

$$E = -\frac{G m_1 m_2}{2a}$$

Same $a \rightarrow$ same energy, regardless of eccentricity.



4. Blackboard Derivation: Vis-Viva Equation

Deriving the Vis-Viva Equation

Goal: Find the speed v at any distance r in an orbit with semi-major axis a .

Starting point: Total energy is conserved and equals $E = -GMm/(2a)$.

At any point on the orbit:

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = -\frac{GMm}{2a}$$

where M is the central mass ($m \ll M$).

Step 1: Evaluate at Perihelion & Aphelion

At perihelion ($r_p = a(1 - e)$) and aphelion ($r_a = a(1 + e)$):

$$\frac{1}{2}mv_p^2 - \frac{GMm}{a(1 - e)} = \frac{1}{2}mv_a^2 - \frac{GMm}{a(1 + e)}$$

Angular momentum conservation:

$$v_p \cdot a(1 - e) = v_a \cdot a(1 + e)$$

$$\Rightarrow v_p/v_a = (1 + e)/(1 - e)$$

Solving yields $E = -GMm/(2a)$

Key insight: Energy depends only on a , **not** on e .

Step 2: The Vis-Viva Equation

Setting instantaneous energy equal to the total:

$$\frac{1}{2}mv^2 - \frac{GMm}{r} = -\frac{GMm}{2a}$$

Dividing by $m/2$ and rearranging:

$$v^2 = GM \left(\frac{2}{r} - \frac{1}{a} \right)$$

Important special cases:

- **Circular orbit** ($r = a$): $v_{\text{circ}} = \sqrt{GM/a}$
- **Escape velocity** ($a \rightarrow \infty$): $v_{\text{esc}} = \sqrt{2GM/r} = \sqrt{2} v_{\text{circ}}$

Application: Earth's Orbital Velocity

Point	Distance r	Velocity
Perihelion	$a(1 - e) = 1.471 \times 10^{11} \text{ m}$	$v_p = 30.3 \text{ km s}^{-1}$
Aphelion	$a(1 + e) = 1.521 \times 10^{11} \text{ m}$	$v_a = 29.3 \text{ km s}^{-1}$
Mean	$a = 1.496 \times 10^{11} \text{ m}$	$\bar{v} = 29.8 \text{ km s}^{-1}$

Nearly circular orbit ($e = 0.017$) $\rightarrow$ only $\pm 0.5 \text{ km s}^{-1}$ variation.

Application: Halley's Comet

Halley: $a = 17.8$ AU, $e = 0.967$ (extreme eccentricity)

Point	Distance	Velocity
Perihelion	0.586 AU	54.6 km s^{-1}
Aphelion	35.1 AU	0.91 km s^{-1}

- Races through the inner solar system at $\sim 2\times$ Earth's speed
- Crawls beyond Neptune at $<1 \text{ km s}^{-1}$
- Same total energy, just redistributed between K and U along the orbit



5. Orbital Resonances & Tidal Forces

Mean-Motion Resonances

- **$p:q$ resonance**: inner body completes p orbits per q of the outer body
- Gravitational kicks add up coherently → can stabilise **or** destabilise

Resonance	Bodies	Effect
3:2	Pluto–Neptune	Protects Pluto from close encounters
Near 5:2	Jupiter–Saturn	“Great inequality”
2:1	Asteroid belt gaps	Kirkwood gaps (destabilising)
1:1	Jupiter Trojans	Co-orbital at L4, L5

The Laplace Resonance

The Galilean moons Io, Europa, Ganymede are in a **1:2:4 period ratio**:

Moon	Period (days)	Ratio to Io
Io	1.769	1
Europa	3.551	≈ 2
Ganymede	7.155	≈ 4

- Maintained by tidal interactions with Jupiter
- **Forces non-zero eccentricities**
- Io's $e \approx 0.004$ drives tidal flexing $\rightarrow$ extreme volcanism (Lecture 3)

Orbital resonances can drive geological activity billions of years after formation.

Secular Resonances: Slow Dynamical Chaos

Beyond mean-motion resonances, orbits are also coupled by **secular** (long-term) precession frequencies.

- A secular resonance occurs when the apsidal or nodal precession frequency of one body matches another's
- In the solar system: ν_5 (Jupiter's apsidal frequency) and ν_6 (Saturn's) resonances define the inner edges of the asteroid belt
- Asteroids near ν_6 at 2.1 AU have their eccentricity pumped until they are ejected or collide with the Sun
- Secular chaos operates on 10^4 – 10^7 yr timescales
- Laskar (1989): Earth's eccentricity evolution is **intrinsically chaotic** beyond ~60 Myr
- Relevance: explains Kirkwood gaps and long-term orbital instabilities

Source: Laskar (1989, 1990); Murray & Dermott (1999)

Tidal Forces

- Gravity varies across an extended body → **differential force**
- Tidal acceleration across a moon of radius R_s at distance d :

$$\Delta a_{\text{tidal}} \approx \frac{2GM_p R_s}{d^3}$$

- Falls off as d^{-3} (steeper than gravity's d^{-2})
- **Tidal bulges:** elongation along the line connecting the bodies
- If rotation $\neq$ orbital period → bulge misalignment → torque
- Energy dissipated as heat → **tidal heating** (Lecture 3)

Tidal Locking (Synchronous Rotation)

- Tidal torque gradually slows rotation until $P_{\text{rot}} = P_{\text{orb}}$
- Same face always points toward the planet
- The Moon is tidally locked to Earth (same face always visible)
- All four Galilean moons are tidally locked to Jupiter
- Locking timescale: $\tau_{\text{lock}} \propto d^6 / (M_p^2 R_s^3)$

Note

Tidal locking is the **equilibrium state**. Tidal heating occurs during the approach to equilibrium (or when resonances maintain eccentricity, preventing full equilibrium).

The Roche Limit

Inside the **Roche limit**, tidal forces exceed self-gravity:

→ satellites are torn apart

For a fluid satellite:

$$d_R \approx 2.46 R_p \left(\frac{\rho_p}{\rho_s} \right)^{1/3}$$

For Saturn + icy satellite:

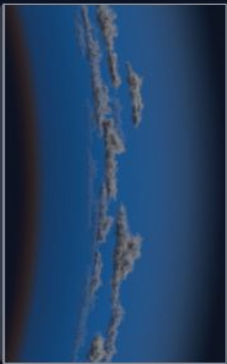
$$d_R \approx 126,000 \text{ km} \approx 2.16 R_{\text{Saturn}}$$

Saturn's main rings extend from 67,000 to 137,000 km, **mostly within the Roche limit** (the formula is approximate; rigid-body and compositional effects shift the boundary).



Credit: NASA/JPL/SSI (Cassini)

6. Planetary Migration



Planet-Disk Interactions

- Planets embedded in a disk excite spiral density waves
- Waves carry angular momentum away from the planet
- Net torque can push the planet **inward or outward**

Type I: Low-mass planet ($\lesssim$ few $M_{\oplus}$), fully embedded
Migrates on $\sim 10^5$ yr timescale (potentially too fast!)

Type II: Massive planet ($\sim M_J$) opens a **gap**
Migrates with the disk on viscous timescale ($\sim 10^5$ – 10^6 yr)

Type III: Rapid runaway migration in massive disks

Lindblad and Corotation Torques

Two kinds of torques act on a planet embedded in a disk:

- **Lindblad torques:** arise at inner and outer Lindblad resonances where spiral density waves form
 - Inner LR exerts a positive torque (outward push)
 - Outer LR exerts a negative torque (inward push)
 - Outer typically wins slightly $\Rightarrow$ inward migration
- **Corotation torques:** arise from material co-orbiting with the planet
 - Depends sensitively on the disk temperature and density gradients
 - Can be positive (outward) and partially balance Lindblad

The real net torque depends on disk structure; this is an active area of research.

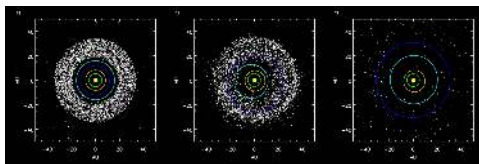
How Does the Gas Disk Go Away?

Planet formation has a hard deadline.

- **Viscous accretion:** gas accretes onto the central star, feeding its growth
- **Photoevaporation:** X-ray and EUV photons from the young star heat the upper disk layers, driving thermal winds
- **MHD winds:** magnetised winds launched from the disk surface carry mass and angular momentum
- Once surface density drops below a threshold, photoevaporation “drains” the remaining gas rapidly
- Total lifetime: 3–5 Myr median (Haisch 2001)

Giant planets must form within 1–3 Myr to capture substantial gas envelopes before the disk disappears.

The Nice Model



Credit: Wikimedia Commons, CC BY-SA 3.0

- Giant planets formed in a **compact** configuration (5–17 AU)
- Jupiter–Saturn crossed 2:1 resonance → **instability**
- Uranus & Neptune scattered outward
- Disrupted the primordial Kuiper Belt
- May explain the Late Heavy Bombardment

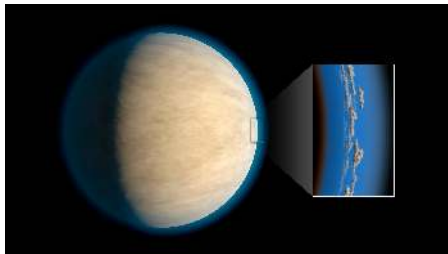
The Grand Tack Hypothesis

1. Jupiter migrated **inward** to ~1.5 AU (Type II migration)
2. Saturn caught up, trapped in resonance
3. Combined torques drove both planets **outward** (“tacking”)
4. Jupiter swept through the inner solar system twice

Potentially explains:

- The small mass of Mars
- The compositional structure of the asteroid belt
- The distribution of water-bearing material in the inner solar system

Evidence for Migration: Hot Jupiters



Credit: ESO/L. Calçada

- Gas giants orbiting at $a \lesssim 0.1$ AU
- Orbital periods of just a few **days**
- Cannot have formed in situ (too hot, too little material)
- Must have formed further out and **migrated inward**

Hot Jupiters are the strongest evidence that planetary migration is real.



AD 209



HD 13006



M148



AD 208



Deneb 24



Deneb 25



Deneb 33



Deneb 39

7. Recent Advances & Outlook

Deneb 37

GM 146

HD 142446

HD 142296



HD 146



M71 146



HD 146



HD 146



Deneb 34



Deneb 35



Deneb 36



High-Eccentricity Migration

Not all migration is disk-driven.

- **Planet-planet scattering:** close encounters between giant planets scatter one inward on a highly eccentric orbit
- **Kozai-Lidov mechanism:** an inclined ($i \gtrsim 40^\circ$) outer companion drives secular oscillations that conserve $L_z = \sqrt{1 - e^2} \cos i$, trading inclination for eccentricity (and back)
- Once eccentric, tidal interactions with the star circularise the orbit at short periods
- Produces hot Jupiters on **high-obliquity** orbits (misaligned with stellar spin)
- Observed: $\sim 1/3$ of hot Jupiters have spin-orbit misalignments $> 30^\circ$

Source: Rasio & Ford (1996); Winn et al. (2010); Dawson & Johnson (2018)

Meteoritic Dichotomy: NC vs. CC

Meteorites divide cleanly into two isotopic groups:

- **Non-carbonaceous (NC):** ordinary, enstatite, and aubrite chondrites; most iron meteorites; originated in the inner solar system
- **Carbonaceous (CC):** CI, CM, CV, CR, CB chondrites; some iron meteorites; originated in the outer solar system
- Distinct ^{50}Ti , ^{54}Cr , ^{48}Ca , ^{94}Mo isotopic signatures
- Implies two separate reservoirs that did not mix for ~3–4 Myr
- Natural explanation: **Jupiter** formed early, blocking inward drift of pebbles

Source: Kruijer et al. (2017), *PNAS*; Warren (2011)

Sample-Return Missions: Accretion Fossils

Recent missions have delivered pristine asteroid samples to Earth.

- **Hayabusa2** (JAXA): returned 5.4 g of asteroid **Ryugu** in 2020
- C-type rubble-pile asteroid, carbonaceous chondrite affinity
- Contains organic molecules including all RNA/DNA bases
- **OSIRIS-REx** (NASA): returned ~70 g of asteroid **Bennu** in 2023
- Hydrated clays, carbonates, phosphates: aqueous alteration
- Both targets formed in the outer main belt; provide snapshots of primordial accretion

Source: Yada et al. (2022) on Ryugu; Lauretta et al. (2024) on Bennu

Juno: Jupiter's Dilute Core

Juno's gravity measurements (Wahl et al. 2017) revolutionised our picture of Jupiter's interior.

- Pre-Juno assumption: compact heavy-element core, pure H/He envelope
- Juno data: the “core” extends to **30–50%** of Jupiter's radius
- Smooth compositional gradient rather than a sharp boundary
- Total heavy-element content: $\sim 20\text{--}40 M_{\oplus}$
- Implication: Jupiter's formation was not a clean core-accretion event
- Possible cause: an early giant impact dispersed the compact core (Liu et al. 2019), or gradual mixing during gas accretion

Source: Wahl et al. (2017); Militzer et al. (2022)

More Recent Advances

- **ALMA MAPS (2021):** chemical maps of 5 disks at planet-forming scales
- **JWST protoplanetary disks:** MIRI detects water and organic molecules in the terrestrial zone of young disks (2023–2024)
- **Pebble accretion:** refined timing constraints on giant-planet formation (<1–3 Myr)
- **Meteorite isotopics:** NC-CC dichotomy + Mo isotopes timing of inner vs. outer disk
- **Psyche:** NASA mission (launched October 2023, arrival 2029) to study a metal-rich asteroid, a possible naked core

JWST: Water and Molecules in Terrestrial Zones

JWST's MIRI instrument can detect molecular emission lines from the inner disks of young stars.

- Detected H_2O , CO_2 , C_2H_2 , HCN , CH_3 in the inner 1–10 AU of several disks
- Water vapour abundance traces ice transport across the snow line
- Pebble drift may enhance inner-disk water content during planet formation
- Evidence that delivery of water to the terrestrial zone is an active, ongoing process during planet formation

Source: Kamp et al. (2023); Banzatti et al. (2023); Perotti et al. (2023)

What Kind of Planets Form?

The outcomes of planet formation, seen in exoplanet surveys:

- **Hot Jupiters:** rare (~1% of solar-type stars), clear evidence for migration
- **Warm/cool gas giants:** ~10% of stars at 1–10 AU
- **Super-Earths / mini-Neptunes:** extremely common, ~30–50% of stars
- **Radius valley:** gap in planet radii around $1.5\text{--}2 R_{\oplus}$ (Lecture 13)
- **Compact multi-planet systems:** many stars host 3–8 planets inside 1 AU
- **Solar-system analogues:** relatively rare

We will return to exoplanet demographics in Lecture 13.

Summary

1. Planets form in **protoplanetary disks** around young stars (lifetime ~3–5 Myr)
2. Dust grows to planetesimals via sticking, streaming instability, and pebble accretion
3. Giant planets form by **core accretion**: solid core → critical mass → runaway gas capture
4. Orbits follow Kepler's laws; the **vis-viva equation** gives speed at any point
5. Orbital **resonances** can protect or destabilise orbits, and drive tidal heating
6. Planets **migrate** through disk and planetesimal interactions, confirmed by hot Jupiters

Questions?

Next: **Lecture 3. Planetary Heat & Energy Transport**

Lecture notes: formingworlds.github.io/IntroductionToPlanetaryScience